

2024

(Session : 2023-26)

(Paper ID : 12007)

Time : 3 hours

Full Marks : 80

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Answer any five questions.

1. (a) Solve the following:

(i) $(41)_{10} = ()_2$

(ii) $(F3A7C2)_{16} = ()_8$

(iii) $110110 + 101$

(iv) $101011 + 111000$

(v) $(623)_8 = ()_2$

(b) Addition of the following:

(i) $1101.11 + 1011.10$

(ii) $10111.01 + 11001.11$

(iii) $100110111 + 10110010$

(iv) $1000011000 + 10000010$

(v) $110100111 + 101100100$

2. (a) Explain the meaning of overflow and underflow with an example.

(b) Define any four number systems depending on different base values. Convert the binary number 11000111.1101 from 2's complement form to equivalent decimal form.

3. Determine by means of a truth table the validity of De-Morgan's theorem for three variables $(ABG)' = A' + B' + C'$.

4. Simplify the following expressions using Boolean Algebra:

(a) $AB + AB'$

(b) $AB + A(CD + CD)$

(c) $(BC' + A'D)(AB' + CD')$

5. Explain Timings sequence digital logic families.

6. Write short notes on the following:

(a) Multiplexer

(b) Disk and Tape

- (c) Magnetic Drum
- (d) Input / Output Interface

7. (a) Give the circuit and working of shift registers. How can we make them shift in both directions?

(b) List any four types of counters with their features.

8. (a) Give the design of any one 4-bit binary decoder.

(b) Explain the T and RS, JK flip-flops with their limitations.

9. What is Counter? Explain Asynchronous counter of MOD 8 with Block diagram and Truth Table.

10. What do you mean by Data Transfer? Differentiate Synchronous and Asynchronous mode of data transfer.

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